

DO GIA HUY

ML / AI Engineer Intern

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📍 Ho Chi Minh City, Vietnam

CAREER OBJECTIVE

AI undergraduate at Swinburne University of Technology with hands-on experience building LLM-powered systems, graph-based ML models, and RAG pipelines, including end-to-end production deployment on AWS – from custom training objectives to cloud-native agentic architectures. Seeking an AI/ML engineering internship to contribute to production-grade intelligent systems.

EXPERIENCE

FCAJ Trainee (AWS × Swinburne) – First Cloud AI Journey Program, Amazon Web Services Vietnam

May – Aug 2026

- Self-directed the design, build, and production AWS deployment of Pedix, a Level-4 agentic RAG pediatric care-navigation app – owned the AI system end-to-end as sole AI/cloud developer.
- Built a custom 5-stage Python reasoning loop with age-stratified retrieval and a 3-layer safety architecture. Deployed across 12 AWS services (EC2, Bedrock, DynamoDB, Cognito, API Gateway, CloudFront) with zero-trust networking and least-privilege IAM roles.

PROJECTS

TOPOLOGY-AWARE LOSS DESIGN FOR DRUG-DRUG INTERACTION PREDICTION (Role: Methodology Developer)

- **Description:** Research project on graph-based DDI severity classification under a strict semi-inductive evaluation setting, where test drugs are topologically isolated – blocking the shortcut exploited by most transductive SOTA models.
 - Designed a composite training objective on top of an R-GCN backbone to handle highly imbalanced graph data, integrating asymmetric cost matrices, cross-space feature alignment, and contrastive learning techniques.
 - Improved Contraindicated-class F1 by ~11% over focal loss baseline on the R-GCN backbone while maintaining higher Macro F1 (+1.1pp) and Accuracy (+0.9pp) overall, with drug features encoded via ChemBERTa (HuggingFace).
- **Tools:** Python, PyTorch, PyTorch Geometric, RDKit, Scikit-Learn, Pandas, Transformers (HuggingFace), Git
- **GitHub:** *[Source code is currently private due to double-blind review. Available upon request for interview purposes]*

PEDIX – AGENTIC RAG PEDIATRIC CARE-PATHWAY NAVIGATOR ON AWS (Role: AI Developer & Cloud Deployment)

- **Description:** Cloud-deployed Level-4 agentic RAG web app guiding parents from a child's symptoms (age 0–5) to the right care pathway – built end-to-end for the Swinburne × AWS First Cloud AI Journey program.
 - Designed and deployed a custom 5-stage agentic RAG reasoning loop (no framework) with age-stratified retrieval in Qdrant and a hybrid 2-layer pediatric safety screen, shipped to production across 12 AWS services with zero-trust networking architecture.
 - Built and ran a 25-case RAG/clinical evaluation harness, achieving 96% keyword retrieval hit-rate@3, 100% age-group filter compliance, and 92% urgency classification within one clinical severity level of ground truth with zero critical safety misses.
- **Tools:** Python, FastAPI, AWS (EC2, Bedrock, DynamoDB, Cognito, API Gateway, CloudFront, S3, ALB), Qdrant, React
- **GitHub:** <https://github.com/Robertise/Pedix>

LLM-BASED AUTONOMOUS AGENTS FOR CYBERSECURITY (5 Members | Role: Project Coordinator & LLM Developer)

- **Description:** A locally deployed multi-agent system for cybersecurity threat analysis and incident response, powered by RAG and LLMs.
 - Built a document ingestion pipeline converting MITRE ATT&CK and AlienVault OTX threat intel into ~300 chunks, then designed the full RAG pipeline (LangChain + Pinecone) with traceable citation outputs.
 - Benchmarked agent responses across 6 threat scenarios – achieving 1.0 precision, 0.75 recall, and 0.85 F1-score.
- **Tools:** Python, LangChain, Pinecone, OpenRouter API, OpenAI API, Docker, Git
- **GitHub:** <https://github.com/Robertise/Cyber-llm-agent>

EDUCATION

Swinburne University of Technology

Expected Graduation: 2027

- Bachelor of Artificial Intelligence – GPA: 3.4 / 4

SKILLS

- **Languages:** English – PTE 62 (~IELTS 6.5) (2024)
- **Programming Languages & Frameworks:** Python, JavaScript
- **AI & Machine Learning:** PyTorch, TensorFlow, LangChain, RAG, Agentic Systems, Prompt Engineering, GNN, Graph ML
- **Cloud & Data:** AWS (EC2, Lambda, Bedrock, DynamoDB, Cognito, API Gateway, CloudFront), PostgreSQL, Qdrant, Pinecone, Docker
- **Soft Skills:** Team Leadership, Self-directed Learning, Problem Solving, Critical Thinking